Class Project: CS428 Design of Secure Protocol

Students are advised to analyse about the following requirement for their project before starting the project.

1. Number of entities involved in secure online (internet) communication for example: role of a user, client, server, cloud server, customer, student, candidate, device like mobile, phone, laptop, car, home appliance (smart TV, Smart fridge etc), RFID reader-tag, sensor or and any other devices which communicate and share information through online mode in your protocol architecture.
2. What are the authentication parameter you are considering i). for user like: id-password, smartcard (RFID card/personal device), Biometric feature of a human, ii) for device for example : device IP address etc.
3. What services you want to provide to end user for example: confidentiality of user and user data (by any encryption/decryption method), integrity of user data (by any hash function/ digital signature method), Authentication of user/device and Authorization to access the service, Non-repudiation (digital signature).
4. Test the security of your protocol by applying any attack model.

Title 1. Design of secure user authentication protocol for smart door lock system for your classroom.

Title 2. Design electronic voting scheme for selection of your class CR.

Title 3. Design a secure protocol for access control of institute exam and result database.

Title 4. Design a secure protocol for vehicular-to-vehicular communication.

Title 5. Design a secure protocol for securely storing and sharing patient data in hospital server.

Title 6. Design a secure protocol for authentication between drone-to-drone communication.

Title 7. Design a secure protocol for supply chain management system

Title 8. Design a secure protocol for smart home system.

Title 9. Design a secure protocol for smart farming.

Title 10. Design a secure protocol for blockchain network.